

# LAB 12: Fisher Information

Q: How does randomness in data  $\rightarrow$  uncertainty in parameter estimation?

How do we know if estimator is most efficient?

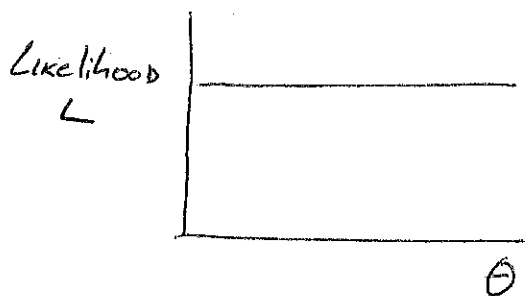
"Fisher information" (Ronald Fisher, 1890-1962)

- Quantifies info that RV 'x' carries about  $M(\theta)$

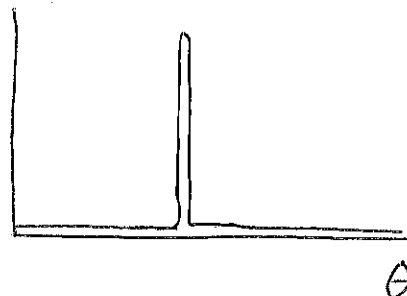
- "The expected value of the observed information"

Model Dep on Parameter(s)  $\theta$

$\Rightarrow$  Intuitive Intro

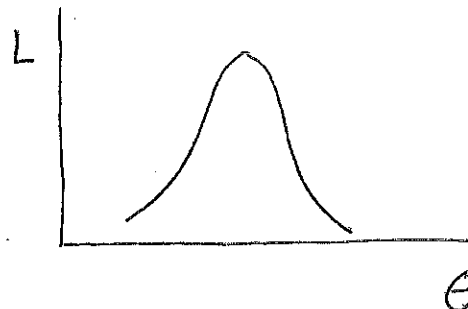


$X_i$  provides no information



$X_i$  provides all necessary info

Most Typical



$X_i$  provides some information

⇒ Intuition (Continued)

- Existence of a slope INDICATES  $X_i$  provides information
- Same thing, because SIGN OF SLOPE DOESN'T MATTER
- Average

$$\left( \frac{\partial}{\partial \theta} \ln(f(x|\theta)) \right)^2 \text{ over PDF}$$

$$\begin{aligned} I &= \int \left( \frac{\partial}{\partial \theta} \ln(f(x|\theta)) \right)^2 f(x|\theta) dx \\ &= E \left[ \left( \frac{\partial}{\partial \theta} \ln(f(x|\theta)) \right)^2 \right] \end{aligned}$$

↑  
Expectation Value

- ↳ The information is large, IF ON AVERAGE  $\ln(f)$  is sensitive to changes in  $\theta$ .

⇒ other facts

$$i) I(\theta) = \langle (l')^2 \rangle = - \langle l'' \rangle \quad \text{Partial Integration}$$

$$ii) I_n(\theta) = n I_1(\theta) \quad \text{Info for } n \text{ measurements}$$

iii) Cramer-Rao Inequality: For UNBIASED ESTIMATORS  
THE VARIANCE OF THE ESTIMATOR CANNOT BE LESS THAN  
THE INVERSE OF FISHER INFORMATION

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)}$$

Ex: Normal Distribution

$$f(x|\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\theta)^2}{2\sigma^2}}$$

$$I(\theta) = \frac{1}{\sigma^2}$$

{ you'll show in lab! }

$$I_N(\theta) = \frac{N}{\sigma^2}$$

$$\text{Var}(\theta) = \frac{\sigma^2}{N}$$

{ Cramer-Rao }

$$\sigma_\theta = \sqrt{\text{Var}(\theta)} \geq \frac{\sigma}{\sqrt{N}}$$

"Error-in-the-mean"  
for Normal Distributions

Ex: Poisson Distribution

$$f(x|\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\ln f = x \ln \lambda - \lambda + \ln(x!)$$

$$\frac{\partial}{\partial \theta} \ln f = \frac{x}{\lambda} - 1$$

$$\frac{\partial^2}{\partial \theta^2} \ln f = -\frac{x}{\lambda^2}$$

$$\hookrightarrow I_x(\lambda) = E \left[ \left( \frac{x}{\lambda} - 1 \right)^2 \right] = E \left[ \frac{x}{\lambda^2} \right] = \frac{1}{\lambda}$$

LAB: Carry out & Plot these steps  
for a Normal Distribution.